Machine Learning and the Robustness of Causal Estimates: Evidence from Re-Estimated Studies in African Development

1 Introduction

Empirical estimates of causal effects often depend on how researchers adjust for observed covariates. Conventional regressions make this adjustment transparent but commonly impose linearity, additivity, and a limited set of researcher-chosen interactions. These restrictions can be consequential when outcomes, treatment assignment, or instrument exposure vary nonlinearly with rich demographic, geographic, and historical information. Simply enlarging the regression is not a general solution: flexible specifications can overfit, become unstable, and make model selection part of the inferential problem. These concerns are especially relevant when the causal parameter is low-dimensional but the conditional expectations used to estimate it are not.

This paper applies double/debiased machine learning (DML) to three influential studies of African political economy and development. The applications cover historical persistence, nation building through shared experiences, and the relationship between taxation and political participation. They also encompass observational regression, instrumental variables, event-timing variation, and clustered random assignment. These settings provide a demanding test bed for flexible adjustment because rich controls and fixed effects coexist with clustered dependence, limited effective sample sizes, and, in the instrumental-variables application, potentially weak residualized first-stage variation. The substantive stakes are equally clear: the original conclusions concern trust, national identity, state capacity, taxation, and citizen engagement.

1.1 Literature Review

Machine learning offers flexible prediction and regularization, yet prediction and causal inference are distinct tasks. A predictor can perform well out of sample without delivering a valid treatment-effect estimate, because causal analysis additionally requires an identified estimand, an appropriate score, and valid uncertainty quantification (Athey and Imbens, 2019). Naively inserting regularized predictions into a causal estimator can transmit shrinkage and overfitting bias to the coefficient of interest. Conversely, causal forests demonstrate that predictive tools can support treatment-effect estimation only after their construction is adapted to causal targets and inference (Wager and Athey, 2018). The relevant question is therefore not whether machine learning predicts well in the abstract, but whether it improves nuisance-function adjustment while respecting the underlying research design.

DML addresses this problem by combining Neyman-orthogonal scores with cross-fitting (Chernozhukov et al., 2018). Orthogonality makes the target score locally insensitive to small errors in estimated nuisance functions, while cross-fitting trains those functions on one sample partition and evaluates the score on held-out observations. Under the required moment, convergence-rate, and identification conditions, this construction can remove first-order regularization bias and support approximately root-n, asymptotically normal inference. The framework accommodates partially linear regression, partially linear instrumental variables, and interactive-regression or augmented inverse-probability scores. It does not, however, create identification: conditional exogeneity, instrument exclusion and relevance, or randomized assignment must still come from the original research design.

Baiardi and Naghi (2024) demonstrate the value of using causal machine-learning methods to revisit published empirical studies. Their evidence suggests that flexible nuisance estimation can reveal nonlinearities, reduce reliance on discretionary specification choices, and assess the stability of established results. At the same time, re-estimation cannot establish whether an original identifying assumption is correct, and differences across estimators need not imply that one estimate is closer to the true causal effect. Building on this measured approach, the present paper shifts attention to prominent applications from Sub-Saharan Africa and compares DML performance across several distinct identification strategies. This cross-design perspective helps separate general lessons about flexible adjustment from problems specific to a particular application.

1.2 Contribution of This Paper

This paper asks: when the identifying design, sample, treatment, outcomes, controls, fixed effects, and clustering choices are held fixed, do DML estimates reinforce, qualify, or weaken the conclusions of influential applied studies? The first application revisits Nunn and Wantchekon's (2011) evidence linking an ethnic group's historical slave exports to lower present-day trust in Africa, re-estimating both the controlled association and the instrumental-variables design based on historical distance to the coast. The second revisits the event-timing estimates of Depetris-Chauvin, Durante, and Campante (2020), which relate national-team football victories to national identification and intergroup attitudes while absorbing country-match, language-year, and calendar fixed effects. The third revisits Weigel's (2020) clustered field experiment in Kananga, Democratic Republic of the Congo, where neighborhoods were assigned to a door-to-door property-tax campaign and subsequent political participation was measured.

The paper's principal contribution is a design-preserving comparison of what changes when nuisance functions are estimated flexibly. In every application, machine learning is confined to nuisance adjustment; the original assignment mechanism, fixed effects, instrument, estimand, and clustering level remain the source of causal interpretation. DML estimates are compared with the published benchmarks across prespecified learners and repeated sample splits, with cluster-aware folds used where observation-level splitting may be optimistic. The analysis evaluates nuisance prediction alongside treatment-residual variation, split-to-split stability, and the precision of the final score. For the instrumental-variables application, particular attention is given to the residualized first stage and score denominator. These diagnostics prevent predictive accuracy from being treated as evidence of causal validity and reveal cases in which flexible learners leave insufficient identifying variation.

The findings are deliberately setting-dependent. Flexible adjustment leaves the football estimates close to their original pattern and broadly preserves the positive participation response to the tax campaign, although evidence is weaker for one participation component after multiplicity-aware inference. In the slave-trade application, partially linear estimates consistently support a negative conditional association, whereas the IV results depend materially on learner choice, fold construction, and residualized first-stage strength. These findings neither validate nor overturn the original identifying assumptions. Instead, they identify where published conclusions survive a common flexible-adjustment exercise and where uncertainty shifts to a design-specific bottleneck. The analysis introduces no new instruments, controls, or sources of quasi-experimental variation, and it uses forests for nuisance estimation rather than heterogeneous-effect discovery.

1.3 Machine-Learning Learners

Across the applications, machine-learning algorithms estimate nuisance functions such as conditional outcomes, treatments, and, in the instrumental-variables design, instruments. They change how observed covariates enter those predictions; they do not replace the original source of identification or determine the causal estimand. Table 1 summarizes the seven learner families reported in the empirical comparisons and the fixed training settings used in the main notebooks.

Table 1: Machine-learning learners used for nuisance estimation

Learner	What it is and does	Functional form	Main training settings
Lasso	An L_1-penalized least-squares regression that shrinks coefficients and can set some exactly to zero (Tibshirani, 1996).	Linear and additive in the supplied features.	Penalty $\alpha = 0.001$, an intercept, and at most 10,000 iterations. The Table 6 implementation standardizes inputs internally.
Elastic net	A regularized regression combining L_1 selection with L_2 shrinkage, which is useful when predictors are correlated (Zou and Hastie, 2005).	Linear and additive in the supplied features.	Campante: $\alpha = 0.001$, L_1 ratio 0.5, and 10,000 iterations. Weigel's prespecified final learner uses standardized or imputed inputs, $\alpha = 0.03$, L_1 ratio 0.5, and 20,000 iterations.

Continued on next page

Table 1 (continued)

Learner	What it is and does	Functional form	Main training settings
Boosting (XGBoost)	A sequence of shallow regression trees, each added to reduce errors left by the preceding ensemble (Friedman, 2001; Chen and Guestrin, 2016).	Nonlinear; learns thresholds and interactions.	300 trees, learning rate 0.03, depth 2, row and feature subsampling of 0.8, and a squared-error objective.
Single tree	A regression tree that recursively partitions the covariate space and assigns a constant prediction within each terminal leaf (Breiman et al., 1984).	Nonlinear; learns threshold effects and interactions.	Maximum depth 8 and minimum leaf size 5. It is one tree, not a forest.
Random forest	An average of many randomized regression trees, using feature subsampling to reduce correlation among trees (Breiman, 2001).	Nonlinear; averages many threshold-and-interaction rules.	500 trees, maximum depth 16, minimum leaf size 5, and square-root feature subsampling.
Neural network	A one-hidden-layer feed-forward network trained by back-propagation to combine weighted nonlinear transformations of the covariates (Rumelhart et al., 1986).	Nonlinear; can learn smooth interactions and transformations.	Standardized inputs; 16 ReLU units; Adam learning rate 0.001; weight decay 0.01; batch size 1,024; at most 250 epochs; and early-stopping patience 20.
Ensemble	An equal-weight arithmetic average of predictions from lasso, XGBoost, random forest, and the neural network (Dietterich, 2000).	Nonlinear overall because it averages linear and nonlinear components.	The four component learners use the settings above. No stacking regression or learned meta-model is fitted.

These classifications concern the nuisance regressions: lasso and elastic net remain linear in the features supplied to them, whereas trees, forests, boosting, and the neural network can learn nonlinearities or interactions without entering each term by hand. The nonlinear learners are not assumption-free; their depth, leaf size, learning rate, architecture, regularization, and stopping rules control flexibility. Cross-fitting and repeated splitting are separate DML procedures rather than learner hyperparameters. Finally, the report's "Best" column is a nuisance-performance and diagnostic selection rule, not another learner and not a search for the preferred treatment-effect sign or significance level.

2 Nunn and Wantchekon: Slave-Trade Exposure and Trust

2.1 Original Analysis

Nunn and Wantchekon (2011) study whether historical exposure to Africa's slave trades helps explain contemporary differences in interpersonal and institutional trust. Their analysis combines individual-level Afrobarometer survey responses with historical data on slave exports by ethnic group. The treatment variable in the specifications revisited here is ancestral exposure to the slave trades, measured as log slave exports normalized by land area. The original analysis estimates ordinary least squares (OLS) regressions of each trust outcome on slave-trade exposure and the original control set. The outcome-specific samples range from 15,905 to 16,709 observations. Inference in the original specification clusters standard errors by ethnicity and district, a two-way clustering structure that is preserved in the re-estimation. The original findings coefficients are negative for all five trust outcomes and statistically significant at the 1 percent level.

The paper also extends the analysis by using an instrumental variables (IV) strategy (Table 6). Slave-trade exposure is instrumented with historical distance from the coast. The identifying argument is that distance from the coast predicts exposure to the transatlantic and Indian Ocean slave trades, while, conditional on the historical and geographic controls, distance should affect modern trust only through slave-trade exposure. The Table 6 IV estimates are also negative across outcomes and similar in magnitude to the original findings.

Following the revisited-study structure in Baiardi and Naghi (2024), the DML exercise below does not introduce a new historical identification strategy. Instead, it keeps the Nunn and Wantchekon estimands fixed and asks whether the original OLS and IV results survive flexible, cross-fitted nuisance adjustment. This is the value added of DML in this setting: the outcome, treatment, and, in the IV case, instrument relationships where the controls can be learned flexibly, while the final coefficient remains an interpretable treatment-effect parameter.

2.2 DML Re-estimation of Table 3

The re-estimation strategy for the orignal findings uses a partially linear DML model, in the form of Chernozhukov et al. (2018). For each trust outcome, the model is

$$Y_i = \theta D_i + g_0(X_i) + u_i, \qquad D_i = m_0(X_i) + v_i, \tag{1}$$

where Y_i is a trust outcome, D_i is slave-trade exposure, and X_i is the original control set. Machine learning is used to estimate the nuisance functions $g_0(X_i)$ and $m_0(X_i)$. The treatment coefficient θ is then estimated by regressing the cross-fitted outcome residual on the cross-fitted treatment residual with no intercept.

We use six learners to estimate the nuisance functions which have been chosen to both replicate the approach of Baiardi and Naghi (2024), and provide a diverse set of learners to estimate the nuisance functions. We use a cross-fitting algorithm which is repeated 100 times with two random folds. In each split, the nuisance learner is trained on one fold and predicts the outcome and treatment on the held-out fold. The out-of-fold predictions are used to form residuals, and the final residual-on-residual coefficient is computed for that split. We then report the median coefficient across splits and use split-adjusted uncertainty. Standard errors are two-way clustered by Murdock ethnicity group and district, matching the original clustered-inference structure and following the logic of two-way clustered covariance estimation in Cameron, Gelbach, and Miller (2011).

The "Best" column in Table 1 is selected by nuisance prediction performance rather than by the sign, size, or statistical significance of the treatment coefficient. The rule ranks learners by outcome-nuisance and treatment-nuisance mean squared error. This column is useful as a compact diagnostic, but it should not be read mechanically as the preferred economic estimate. A learner can predict the treatment well while leaving a residualized treatment variation that changes the second-stage coefficient substantially.

Table 2: DML re-estimates of Nunn and Wantchekon Table 3

	(1) Original OLS	(2) Lasso	(3) Boosting	(4) Ensemble	(5) Neural Net	(6) Tree	(7) Forest	(8) Best
Panel A: DML estimates								
Trust relatives	-0.178	-0.184	-0.082	-0.185	-0.177	-0.060	-0.026	-0.026
	[0.032]	[0.035]	[0.064]	[0.058]	[0.062]	[0.114]	[0.073]	[0.073]
Trust neighbors	-0.202	-0.203	-0.107	-0.216	-0.213	0.053	-0.066	-0.066
	[0.031]	[0.032]	[0.068]	[0.057]	[0.069]	[0.147]	[0.086]	[0.086]
Trust local council	-0.129	-0.121	-0.119	-0.136	-0.128	0.095	0.023	0.023
	[0.021]	[0.023]	[0.063]	[0.045]	[0.064]	[0.156]	[0.073]	[0.073]
Intragroup trust	-0.188	-0.195	-0.104	-0.199	-0.181	0.141	0.003	0.003
	[0.032]	[0.032]	[0.068]	[0.059]	[0.074]	[0.152]	[0.090]	[0.090]
Intergroup trust	-0.115	-0.117	0.027	-0.107	-0.159	0.281	0.044	0.044
	[0.030]	[0.031]	[0.070]	[0.057]	[0.075]	[0.147]	[0.088]	[0.088]
Observations, range	15,905–16,709							
Ethnicity clusters, range	146–147							
District clusters, range	1,184–1,194							
Repeated splits	100							
Cross-fitting folds	2							

Notes: Original column reports the Nunn and Wantchekon Table 3 OLS coefficient and two-way clustered standard error from the replication log. DML columns report repeated cross-fit partially linear estimates. Main DML cross-fitting uses random observation-level folds. Standard errors are in brackets. The Best column follows the notebook nuisance-prediction selection rule, including first-stage and denominator penalties in IV tables.

Table 1 compares the original Table 3 OLS coefficients with the main random-fold DML estimates. The central result is that stable regularized and ensemble-style learners preserve the original negative relationship between slave-trade exposure and trust. Lasso is especially close to the original OLS coefficients.

The ensemble and neural-network estimates also support the original sign pattern, remaining negative for all five outcomes. These estimates indicate that once the nuisance functions are estimated with regularized, smooth, or averaged learners, the DML exercise largely reinforces the original Table 3 finding.

The more flexible tree-based learners are less stable. Boosting attenuates several estimates and reverses the sign for intergroup trust. The single tree and random forest estimates are often much closer to zero, and several become positive. The tree estimate is positive for trust in neighbors, trust in the local council, intragroup trust, and intergroup trust; the forest estimate is positive for trust in the local council, intragroup trust, and intergroup trust. This does not overturn the original result, but it shows that the magnitude and, for some learners, the sign can be sensitive to how the nuisance functions are learned.

2.3 Robustness and Sensitivity

Due to the lack of stability in some of the learners, primarily Tree and Forest learners, we replace the random observation-level folds with ethnicity-cluster folds. Instead of allowing observations from the same ethnicity group to appear in both training and held-out folds, this design holds out the entire Murdock ethnicity groups. We suspect that the flexible learners, especially Tree and Forest methods, may learn ethnicity-level exposure patterns from other respondents in the same ethnicity.

Table 2 reports the ethnicity-cluster cross-fitting results. This design generally attenuates the DML estimates and increases uncertainty, which is expected because treatment and outcome prediction become harder when entire ethnic groups are excluded from training folds. Importantly, however, the sign pattern becomes more consistently negative than in the random-fold "Best" column. Lasso remains negative for all five outcomes, with estimates ranging from -0.176 for intragroup trust to -0.102 for intergroup trust. Boosting and the ensemble are also negative across all outcomes. The cluster-cross-fit results therefore improve the consistency of all learners, with Forest and Tree learners becoming more stable and consistent with the other learners

Table 3: DML re-estimates of Nunn and Wantchekon Table 3 with ethnicity-cluster cross-fitting

	(1) Original OLS	(2) Lasso	(3) Boosting	(4) Ensemble	(5) Neural Net	(6) Tree	(7) Forest	(8) Best
Panel A: DML estimates with ethnicity-cluster cross-fitting								
Trust relatives	-0.178	-0.144	-0.107	-0.133	-0.118	-0.048	-0.112	-0.112
	[0.032]	[0.062]	[0.058]	[0.058]	[0.085]	[0.101]	[0.058]	[0.058]
Trust neighbors	-0.202	-0.171	-0.132	-0.163	-0.156	-0.099	-0.139	-0.139
	[0.031]	[0.055]	[0.048]	[0.044]	[0.065]	[0.081]	[0.049]	[0.049]
Trust local council	-0.129	-0.107	-0.084	-0.091	-0.070	-0.093	-0.073	-0.091
	[0.021]	[0.035]	[0.031]	[0.035]	[0.064]	[0.061]	[0.036]	[0.035]
Intragroup trust	-0.188	-0.176	-0.126	-0.151	-0.129	-0.104	-0.125	-0.125
	[0.032]	[0.052]	[0.048]	[0.044]	[0.067]	[0.094]	[0.048]	[0.048]
Intergroup trust	-0.115	-0.102	-0.063	-0.089	-0.100	-0.053	-0.070	-0.063
	[0.030]	[0.041]	[0.040]	[0.036]	[0.057]	[0.071]	[0.039]	[0.040]
Split cluster	–	murdock_name	murdock_name	murdock_name	murdock_name	murdock_name	murdock_name	murdock_name
Observations, range				15,905–16,709				
Ethnicity clusters, range				146–147				
District clusters, range				1,184–1,194				
Repeated splits				100				
Cross-fitting folds				2				

Notes: Original column reports the Nunn and Wantchekon Table 3 OLS coefficient and two-way clustered standard error from the replication log. DML columns report repeated cross-fit partially linear estimates. Cluster-cross-fitting holds out entire Murdock ethnicity groups in each fold. Standard errors are in brackets. The Best column follows the notebook nuisance-prediction selection rule, including first-stage and denominator penalties in IV tables.

We also perform a sensitivity analysis to determine whether the chosen hyper-parameters heavily impact the reported coefficients. Multiple conservative learners are fit to explore whether the positive Tree and Forest estimates were caused by overly flexible learners. The learners used include: a stronger lasso, elastic net, ridge-style elastic net, shallower trees, shallower forests with larger leaves, a smaller regularized neural network, and a conservative ensemble. Then, we run both a random-fold conservative DML and an ethnicity-cluster 5-fold conservative DML.

The conservative sensitivity results strongly clarify the Table 3 interpretation. Under the ethnicity-cluster 5-fold conservative design, all 55 learner-by-outcome estimates are negative. The notebook's nuisance-prediction rule selects the conservative ensemble for all five outcomes. These estimates are smaller in magnitude than the original OLS estimates, but they preserve the same substantive conclusion: greater ancestral slave-trade exposure is associated with lower contemporary trust.

The sensitivity exercise also explains the earlier tree and forest behavior. Conservative tree and forest settings reduce learner flexibility by limiting depth and requiring larger leaves. Once this is done, forests become more consistent with the negative estimates from the regularized learners, especially under ethnicity-cluster folds. Single trees remain less precise and show more split variability, which is unsurprising because a single tree is a high-variance learner. The best reading is therefore that Table 3 is robust in direction, but not in exact magnitude. The negative relationship survives regularized, ensemble, cluster, and conservative sensitivity checks, while highly flexible tree-based nuisance adjustment should be treated as a warning about residualization sensitivity rather than as a preferred specification.

2.4 DML-IV Re-estimation of Table 6

Nunn and Wantchekon (2011) further test the main findings from Table 3 using an instrumental variable approach. The original methodology instruments slave-trade exposure with historical distance from the coast, and are presented in Table 6 of the original paper. To re-estimate the findings using DML, we preserve not only residualized treatment variation but also a stable residualized first stage. This makes the DML-IV more fragile than the partially linear Table 3 exercise.

Our DML-IV approach residualizes the outcome, treatment, and instrument with respect to the controls. In each split, learners estimate the nuisance functions for $E[Y_i \mid X_i]$, $E[D_i \mid X_i]$, and $E[Z_i \mid X_i]$, where Z_i is the distance-based instrument. The final IV coefficient is then computed from the residualized IV score. The key denominator is the covariance between the residualized treatment and the residualized instrument. If this denominator becomes small or unstable, the DML-IV estimate can become noisy even when the nuisance learner has good predictive performance.

We use the same broad learner set as the Table 3 notebook to ensure consistence between estimates: lasso, boosting, ensemble, neural network, tree, and forest. It again uses 100 repeated two-fold cross-fitting splits, median estimates across splits, split-adjusted uncertainty, and two-way clustered standard errors by ethnicity and district. The best-learner rule is stricter for IV than for Table 3. In addition to nuisance prediction performance, it penalizes weak residualized first stages and unstable denominators. This is essential because IV-DML is informative only if the residualized instrument still predicts the residualized treatment in a stable way.

Table 4: DML-IV re-estimates of Nunn and Wantchekon Table 6

	(1) Original IV	(2) Lasso	(3) Boosting	(4) Ensemble	(5) Neural Net	(6) Tree	(7) Forest	(8) Best
Panel A: DML-IV estimates								
Trust relatives	-0.172	-0.175	-0.194	-0.197	-0.721	-6.255	-0.031	-0.197
	[0.076]	[0.075]	[0.319]	[0.165]	[0.574]	[149.617]	[0.563]	[0.165]
Trust neighbors	-0.271	-0.267	-0.273	-0.329	-0.723	-2.837	-0.324	-0.329
	[0.088]	[0.085]	[0.338]	[0.179]	[0.541]	[90.076]	[0.642]	[0.179]
Trust local council	-0.262	-0.247	-0.473	-0.377	-0.820	-2.017	-0.948	-0.377
	[0.075]	[0.073]	[0.339]	[0.179]	[0.603]	[8.434]	[0.854]	[0.179]
Intragroup trust	-0.254	-0.251	-0.375	-0.373	-0.893	-4.904	-0.705	-0.373
	[0.109]	[0.105]	[0.400]	[0.206]	[0.559]	[72.832]	[0.651]	[0.206]
Intergroup trust	-0.189	-0.187	-0.413	-0.327	-0.782	-2.881	-0.738	-0.327
	[0.103]	[0.100]	[0.409]	[0.211]	[0.595]	[58.362]	[0.753]	[0.211]
Observations, range	15,905–16,709							
Ethnicity clusters, range	146–147							
District clusters, range	1,184–1,194							
Repeated splits	100							
Cross-fitting folds	2							
Median DML first-stage F	–	21.330	10.754	18.226	6.949	0.686	4.934	18.226
Median denominator CV	–	0.006	0.025	0.019	0.104	0.870	0.086	0.019
Weak first-stage outcomes	–	0/5	0/5	0/5	5/5	5/5	5/5	0/5
Unstable denominator outcomes	–	0/5	0/5	0/5	0/5	5/5	0/5	0/5

Notes: Original column reports the Nunn and Wantchekon Table 6 linear IV coefficient and standard error. DML-IV columns report repeated cross-fit partially linear IV estimates. Median DML first-stage F and denominator CV summarize learner diagnostics across the five outcomes; weak first-stage is F $<$ 10 and unstable denominator is CV $>$ 0.25. Main DML cross-fitting uses random observation-level folds. Standard errors are in brackets. The Best column follows the notebook nuisance-prediction selection rule, including first-stage and denominator penalties in IV tables.

Table 3 compares the original Table 6 IV estimates with the main random-fold DML-IV estimates. The strongest result is that lasso almost reproduces the original IV estimates outcome by outcome. These close matches support the direction and approximate magnitude of the original IV findings under a regularized DML-IV implementation.

The ensemble estimates are also negative for all five outcomes, although they are often larger in absolute value than both the original IV and lasso DML-IV estimates. These estimates reinforce the original negative trust relationship, but they also show that flexible nuisance adjustment can affect IV magnitudes even when the sign remains stable.

The neural network, tree, and forest estimates should not be treated as preferred IV estimates. The neural-network estimates are large in absolute value and imprecise. Tree estimates are extreme, including -6.255 for trust in relatives and -4.904 for intragroup trust, with enormous standard errors. Forest estimates are also diagnostically weak. The first-stage and denominator diagnostics explain why. In the main Table 6 results, the median DML first-stage F statistic is strong for lasso (21.330), boosting (10.754), and the ensemble (18.226), but weak for the neural network (6.949), tree (0.686), and forest (4.934). The weak-first-stage count is 0/5 for lasso, boosting, and the ensemble, but 5/5 for the neural network, tree, and forest. The tree learner also has unstable denominators for all five outcomes.

2.5 Table 6 Robustness and Sensitivity

Following a similar approach as the previous re-estimate, we check whether an ethnicity-cluster, in which the we hold out the entire Murdock ethnicity groups during cross-fitting is performed. This is especially demanding in the IV setting because the nuisance models must predict the outcome, treatment, and instrument out of ethnicity group, while the residualized treatment and residualized instrument must still retain a stable first stage. Thus, the check asks not only whether flexible adjustment generalizes across ethnic groups, but also whether the IV score remains informative after that stricter residualization.

Table 5: DML-IV re-estimates of Nunn and Wantchekon Table 6 with ethnicity-cluster cross-fitting

	(1) Original IV	(2) Lasso	(3) Boosting	(4) Ensemble	(5) Neural Net	(6) Tree	(7) Forest	(8) Best
Panel A: DML-IV estimates with ethnicity-cluster cross-fitting								
Trust relatives	-0.172	-0.149	0.018	-0.077	-0.168	0.006	-0.002	-0.149
	[0.076]	[0.148]	[0.196]	[0.128]	[0.360]	[0.772]	[0.137]	[0.148]
Trust neighbors	-0.271	-0.251	-0.044	-0.148	-0.231	-0.017	-0.073	-0.073
	[0.088]	[0.165]	[0.163]	[0.118]	[0.355]	[1.187]	[0.123]	[0.123]
Trust local council	-0.262	-0.137	-0.133	-0.136	-0.190	-0.171	-0.114	-0.137
	[0.075]	[0.157]	[0.147]	[0.120]	[0.244]	[1.039]	[0.115]	[0.157]
Intragroup trust	-0.254	-0.229	-0.085	-0.149	-0.195	-0.040	-0.066	-0.229
	[0.109]	[0.146]	[0.152]	[0.113]	[0.698]	[2.452]	[0.120]	[0.146]
Intergroup trust	-0.189	-0.099	-0.053	-0.072	-0.105	-0.047	-0.005	-0.099
	[0.103]	[0.138]	[0.133]	[0.105]	[0.312]	[62.849]	[0.114]	[0.138]
Split cluster	–	murdock_name	murdock_name	murdock_name	murdock_name	murdock_name	murdock_name	murdock_name
Observations, range				15,905–16,709				
Ethnicity clusters, range				146–147				
District clusters, range				1,184–1,194				
Repeated splits				100				
Cross-fitting folds				2				
Median DML first-stage F	–	10.339	4.933	6.719	3.960	2.202	6.925	10.339
Median denominator CV	–	1.235	0.356	0.333	0.461	0.612	0.262	1.235
Weak first-stage outcomes	–	1/5	5/5	5/5	5/5	5/5	5/5	1/5
Unstable denominator outcomes	–	5/5	5/5	5/5	5/5	5/5	4/5	5/5

Notes: Original column reports the Nunn and Wantchekon Table 6 linear IV coefficient and standard error. DML-IV columns report repeated cross-fit partially linear IV estimates. Median DML first-stage F and denominator CV summarize learner diagnostics across the five outcomes; weak first-stage is F < 10 and unstable denominator is CV > 0.25. Cluster-cross-fitting holds out entire Murdock ethnicity groups in each fold. Standard errors are in brackets. The Best column follows the notebook nuisance-prediction selection rule, including first-stage and denominator penalties in IV tables.

Table 4 shows that ethnicity-cluster cross-fitting substantially weakens the DML-IV design. The lasso estimates remain negative for all five outcomes, but the magnitudes are attenuated and standard errors are larger. Other learners often move close to zero. More importantly, the diagnostics deteriorate sharply. The median denominator coefficient of variation is above the instability threshold for every learner, and weak first-stage warnings appear for most learners and outcomes. Even lasso, the most interpretable learner in the main DML-IV table, has unstable-denominator warnings for all five outcomes under cluster cross-fitting. These results should be read as a stress test rather than as a preferred causal specification. They do not necessarily reject the original IV conclusion, but they show that IV-DML is much more sensitive than DML when nuisance prediction is forced to generalize out of ethnicity group.

A sensitivity analysis is also used to determine whether alterations in each learner can reduce the large error and low first-stage F score seen in the DML-IV. As before, we use a stronger lasso, elastic net, ridge-style elastic net, shallower trees and forests, a conservative neural network, and a conservative ensemble. As done before, the goal is to test whether flexible learner instability can be reduced by making the nuisance models less flexible. The analysis is run under both random folds and ethnicity-cluster 5-fold cross-fitting.

The random-fold conservative DML-IV estimates are directionally consistent with the original IV results. The best estimates are negative for all five outcomes with the selected learner being boosting in each random-fold conservative outcome, and the residualized first-stage F statistics are just above 10. However, the standard errors are large and the estimates are generally not statistically significant. This means the random-fold conservative IV results are supportive in sign, but not strong evidence on their own.

The ethnicity-cluster conservative DML-IV design remains fragile. The selected best estimates are negative for all outcomes, but four of the five best estimates have weak-first-stage flags. The only best estimate without a weak-first-stage flag, trust in the local council, has an unstable-denominator flag. Across all learners, the diagnostics are also poor: the random-fold conservative design has 25 weak-first-stage flags out of 55 estimates and 9 unstable-denominator flags, while the ethnicity-cluster conservative design has 35 weak-first-stage flags, 35 unstable-denominator flags, and 33 high split-variability flags. Conservative settings therefore help Table 3 much more than Table 6. In the IV case, the problem is not only overfitting by flexible learners; it is that the residualized IV first stage often becomes weak once the instrument, treatment, and outcome are all residualized by flexible nuisance functions.

The lesson from Table 6 is therefore more qualified than the lesson from Table 3. Lasso and the ensemble provide meaningful support for the original IV direction in the main random-fold DML-IV run. But the cluster and conservative sensitivity checks show that the IV-DML design is fragile when evaluated through first-stage and denominator diagnostics. Tree and forest learners should not be used as preferred Table 6 DML-IV estimates, and even conservative IV-DML results should be interpreted through the diagnostic flags.

2.6 Conclusion

The DML exercise adds three main pieces of evidence to the original Nunn and Wantchekon analysis. First, the Table 3 partially linear DML results strongly support the original negative association between ancestral slave-trade exposure and contemporary trust under stable learners. Lasso nearly reproduces the original OLS coefficients, while the ensemble and neural network also preserve negative signs across all five outcomes. Second, the Table 3 robustness checks show that the negative relationship survives ethnicity-cluster cross-fitting and conservative learner sensitivity analysis, although the magnitudes are generally smaller. Third, the Table 6 DML-IV results are directionally supportive under lasso and the ensemble, but much more diagnostically fragile because IV-DML requires a stable residualized first stage.

DML mostly reinforces the original negative relationship while clarifying where the evidence is strongest. The strongest machine-learning re-estimation evidence is the partially linear DML analysis, especially under regularized, ensemble, and conservative cluster-cross-fit learners. The DML-IV analysis supports the original IV direction under stable learners, but suffers due to weak first-stage and unstable-denominator causing unstable insignificant results.

3 Campante et al.: Building Nations Through Shared Experiences

3.1 Original Analysis

Campante et al. study whether shared national experiences can affect national identity and intergroup attitudes. Their empirical setting uses national football matches as short-run shared events and compares Afrobarometer respondents interviewed shortly before and shortly after national-team victories. The treatment in the specifications revisited here is `post_victory`, an indicator equal to one for respondents interviewed after a national-team victory. The DML re-estimation focuses on the paper's main individual-level Afrobarometer results: Table 2 column 6, ethnic over national identity; Table 4 column 1, trust in countrymen; Table 4 column 2, interethnic trust; Table 4 column 3, liking neighbors from other ethnicities; and Table 4 column 4, dislike of foreign neighbors.

The original identification strategy is an event-timing fixed-effect design. Respondents interviewed near the same national football event are compared before and after the victory. Country-match fixed effects anchor the comparison around the same event, while language-year fixed effects and date-related fixed effects absorb language-group and interview-timing patterns. The core specifications also control for male, age, age squared, unemployment, rural residence, and education. The fixed effects are country-match, language-year, day-of-week, day, and month. Standard errors are clustered by `country_year_fe`.

The published estimates indicate that national football victories reduce ethnic-over-national identification and increase several measures of intergroup trust. The Table 2 ethnic-identity estimate is -0.053 with a standard error of 0.016 and 37,060 observations. In Table 4, the published estimates are 0.063 for trust in countrymen, 0.140 for interethnic trust, 0.102 for liking neighbors from other ethnicities, and 0.019 for dislike of foreign neighbors. The first three Table 4 effects are positive and statistically meaningful in the original analysis, while the dislike-of-foreign-neighbors estimate is small and imprecise. The DML exercise therefore asks whether the main identity and trust findings survive flexible nuisance adjustment while preserving the original event-timing design.

3.2 DML Re-estimation Strategy

To implement the DML estimation strategy in this context we use a fixed-effect residualization where we first partial out the original fixed effects, then run the DML on the fixed-effect residualzied outcome, treatment and controls. This is necessary since the fixed effects can not be freely learned or omitted like the ordinary controls.

The original fixed-effect regression is

$$Y_i = \theta D_i + X_i'\beta + \alpha_{cm(i)} + \lambda_{\ell y(i)} + \delta_{dow(i)} + \eta_{day(i)} + \mu_{month(i)} + \varepsilon_i, \tag{2}$$

where D_i is `post_victory`, X_i is the respondent control vector, and the remaining terms are the original fixed effects. The DML version uses fixed-effect-residualized variables and estimates

$$Y_i^* = \theta D_i^* + g_0(X_i^*) + u_i, \qquad D_i^* = m_0(X_i^*) + v_i. \tag{3}$$

The final coefficient is obtained from the residualized outcome and residualized treatment after cross-fitted nuisance prediction.

The learner set includes lasso, elastic net, boosting, random forest, tree, neural net, and an ensemble. The full run uses 100 repeated two-fold cross-fitting splits, and inference is clustered by country-year to match the original inference level. The best learner for each outcome is selected using rank score over outcome nuisance MSE, treatment nuisance MSE, and split stability. This rule is useful for summarizing nuisance performance, but is not to be interpreted as a ranking of causal estimates.

3.3 Main Results: Original Estimates, Replicated OLS, and Best DML

Table 6 reports the published estimates, the replicated fixed-effect OLS estimates, and the best DML estimates. The DML estimates track the published and replicated fixed-effect estimates for the first four outcomes closely. This is important because the re-estimation preserves the original sample sizes and fixed-effect structure, so the comparison is made against the same substantive estimand.

Table 6: Replicated fixed-effect OLS and best DML re-estimates of Campante et al. Table 2 and Table 4

Outcome	Original table	N	Clusters	(1) Published	(2) Rep. OLS	(3) Best DML	(4) Best learner	(5) OLS q	(6) DML q
Ethnic identity	Table 2 col. 6	37,060	57	-0.053	-0.053	-0.052	Ensemble	0.002	0.003
				[0.016]	[0.016]	[0.017]			
Trust countrymen	Table 4 col. 1	9,355	13	0.063	0.063	0.063	Elastic Net	0.004	0.003
				[0.021]	[0.021]	[0.021]			
Interethnic trust	Table 4 col. 2	7,973	8	0.140	0.143	0.143	Lasso	< 0.001	< 0.001
				[0.040]	[0.038]	[0.037]			
Like other groups	Table 4 col. 3	7,511	12	0.102	0.102	0.101	Lasso	0.002	0.003
				[0.030]	[0.030]	[0.032]			
Dislike foreigners	Table 4 col. 4	7,497	12	0.019	0.019	0.018	Lasso	0.277	0.306
				[0.018]	[0.018]	[0.018]			
Repeated splits	100								
Cross-fitting folds	2								
Treatment	Post-victory indicator								
Fixed effects	Country-match, language-year, day-of-week, day, and month								

Notes: Published estimates are the paper's Table 2 column 6 and Table 4 columns 1-4 benchmarks. Rep. OLS is the notebook's fixed-effect replication with standard errors clustered by country-year. Best DML uses the notebook's nuisance-prediction selection rule after fixed-effect residualization. Standard errors are in brackets. OLS q and DML q are Benjamini-Hochberg adjusted q-values across the five reported outcomes. The full run used 100 repeated two-fold cross-fitting splits on a Colab CUDA runtime.

All 5 outcome groups produce a near identical estimate in both the replicated OLS and best DML learner measurements. This results is not surprising, since both estimators use exactly the same identifying structure and clustering. Due to the structure of the original analysis, the fixed effects absorb most of the systematic treatment variation so after residualization, respondent characteristics have almost no additional ability to predict interview timing relative to a victory.

The similarity in results indicates that, conditional on the fixed effects, post-victory exposure is nearly orthogonal to the observed respondent controls, hence DML finds little additional selection to remove. The original control set is also small and relatively simple, including sex, age, age squared, unemployment, rural status, and education. This provides a limited scope for flexible learners to uncover major nonlinearities or interactions that alter treatment coefficients heavily.

3.4 Learner Sensitivity

Table 10 compares the DML estimates across all learner families. Regularized linear learners, especially lasso and elastic net, closely match the original fixed-effect estimates for each outcome group. The ensemble also tracks the original results for the main identity and trust outcomes. Neural net estimates are generally close to the regularized learners. As seen in the re-estimate of Nunn and Wantchekon (2011), the Tree and Forest methods are less stable, especially in smaller sample groups. This is to be expected since flexible partition-based learners generally have difficulty in samples with few treated observations, limited effective overlap, and only a small number of clusters.

Table 7: DML learner sensitivity for Campante et al. Table 2 and Table 4 outcomes

Outcome	Original	Rep. OLS	Lasso	Elastic Net	Boosting	Forest	Tree	Neural Net	Ensemble
Ethnic identity	-0.053	-0.053	-0.052	-0.053	-0.052	-0.048	-0.046	-0.052	-0.052
	[0.016]	[0.016]	[0.016]	[0.016]	[0.017]	[0.018]	[0.018]	[0.016]	[0.017]
Trust countrymen	0.063	0.063	0.063	0.063	0.060	0.050	0.036	0.067	0.061
	[0.021]	[0.021]	[0.020]	[0.021]	[0.023]	[0.032]	[0.040]	[0.022]	[0.023]
Interethnic trust	0.140	0.143	0.143	0.143	0.145	0.146	0.118	0.144	0.145
	[0.040]	[0.038]	[0.037]	[0.037]	[0.051]	[0.054]	[0.059]	[0.038]	[0.045]
Like other groups	0.102	0.102	0.101	0.102	0.105	0.096	0.088	0.103	0.104
	[0.030]	[0.030]	[0.032]	[0.032]	[0.039]	[0.041]	[0.058]	[0.032]	[0.035]
Dislike foreigners	0.019	0.019	0.018	0.019	0.032	0.042	0.038	0.017	0.027
	[0.018]	[0.018]	[0.018]	[0.018]	[0.022]	[0.024]	[0.041]	[0.020]	[0.017]
Observations, range					7,497–37,060				
Country-year clusters, range					8–57				
Repeated splits					100				
Cross-fitting folds					2				

Notes: DML columns report fixed-effect residualized partially linear estimates for each learner. Forest denotes random forest. Standard errors are in brackets and use the notebook's country-year clustered inference. Learner sensitivity is most visible for tree and forest learners in the smaller Table 4 samples; the linear/regularized learners and ensemble generally track the published fixed-effect estimates closely.

The instability of the foreign-neighbor outcome is particularly revealing. Because its original signal is weak, different nuisance algorithms can move the coefficient substantially in proportional terms even though none provides compelling evidence of an effect.

3.5 Conclusion

Our analysis indicate that the paper's main findings are robust to the functional form used for the observed covariates. The limited movement in the treatment estimates therefore indicates that omitted nonlinearities and interactions among these measured characteristics are unlikely to explain the original estimates.

This is reinforced by the treatment-nuisance diagnostics. For most outcomes, the standard deviation of residualized treatment barely changes after predicting treatment from the residualized controls. The controls consequently contain little incremental information about post-victory interview status once the event-timing fixed effects have been removed. This is an informative covariate-balance result: it suggests that the identifying variation is primarily generated by the timing design rather than by observable respondent composition.

The repeated-cross-fitting results are also stable for the regularized learners. Split-to-split standard deviations are generally small for Lasso, Elastic Net, and the selected specifications. Thus, the principal estimates do not appear to be artifacts of one favorable train-test partition.

While the results address the observed-covariate functional form, we do not test the validity of the quasi-experimental design. It is important to note DML cannot test the structure design of the original experiment, which includes the survey timing around football matches being exogenous, whether survey fieldwork was rescheduled following victories, or whether response and interview completion differ before and after matches.

We also include only the set of controls given in the original paper. Since this is relatively small, our report is not a broad high-dimensional confounding exercise, including other controls could theoretically strength the analysis.

4 Weigel: The Participation Dividend of Taxation

4.1 Original Analysis

Weigel studies whether a state-led property-tax campaign generated a "participation dividend" by increasing citizens' civic and political participation. The empirical setting is a randomized tax intervention in Kananga, Democratic Republic of Congo. Neighborhoods were assigned to a bundled property-tax campaign that included property registration, taxpayer identification, solicitation of payment, and related collection procedures. The target estimand for the re-estimation is therefore the intention-to-treat effect of assignment to the tax campaign, not the effect of payment, registration, visits, or receipt status. The identifying variation comes from the randomized neighborhood-level assignment within strata.

The DML exercise revisits the paper's Table IV participation outcomes. These outcomes are town hall attendance, evaluation-form submission, an indicator for either town hall attendance or evaluation submission, an indicator for both town hall attendance and evaluation submission, a standardized participation index, cost participation as a relative weighted outcome, and a second cost-participation measure that adds opportunity costs. Outcomes are measured at the individual level, but treatment is assigned at the neighborhood cluster level. The original analysis reports OLS-style treatment effects with controls, stratum structure, cluster-robust standard errors, and randomization-inference p-values.

The published Table IV estimates imply a positive participation dividend. Assignment to the tax campaign increased town hall attendance by 0.045 percentage points, with a standard error of 0.020 and a randomization-inference p-value of 0.023. The estimate for evaluation submission is 0.024, with a standard error of 0.012 and a randomization-inference p-value of 0.058. The broader participation outcomes are also positive: 0.050 for town hall or evaluation, 0.027 for town hall and evaluation, 0.145 for the participation index, 0.050 for cost participation, and 0.071 for cost participation 2. The original interpretation is that the tax campaign did not merely increase fiscal contact with the state; it also increased participation in civic opportunities connected to the campaign.

4.2 Design-Aware DML Methodology

The re-estimation preserves Weigel's experimental source of identification while replacing conventional linear covariate adjustment with cross-fitted nuisance learning and Neyman-orthogonal scores (Weigel, 2020; Chernozhukov et al., 2018). Let i index respondents, c neighborhood clusters, and $S_c = s$ the randomization stratum containing neighborhood c. Let $D_c \in \{0, 1\}$ indicate assignment of neighborhood c to the bundled tax campaign, Y_{ic} denote one of the seven Table IV participation outcomes, and X_{ic} collect the verified pre-treatment nuisance covariates. In this notation, the original OLS benchmark can be written as

$$Y_{ics} = \beta D_c + \alpha_s + X_{ic}'\gamma + X_c'\delta + \varepsilon_{ics}, \tag{4}$$

where α_s represents stratum effects and inference is clustered at the neighborhood level. The target remains the reduced-form intention-to-treat (ITT) effect of assignment to the campaign. Neither DML route estimates the effect of registering, paying, receiving a collector visit, or holding a receipt.

Because assignment occurred at the neighborhood level within 33 strata, the treatment probability is a design quantity rather than an observational propensity to be learned from respondent characteristics. For stratum s, the notebook reconstructs it from the complete assignment frame as

$$\widehat{p}_s = \frac{\#\{c : S_c = s, D_c = 1\}}{\#\{c : S_c = s\}}, \qquad 0 < \widehat{p}_s < 1. \tag{5}$$

This probability is mapped to every respondent in the corresponding stratum and audited before estimation. Outcomes use locked, outcome-specific samples. Town hall attendance contains 1,934 respondents in 252 neighborhoods because respondents sampled after meetings were discontinued did not have the same opportunity to attend; the other Table IV outcomes contain 2,913 respondents in 356 neighborhoods.

DML enters through three linked operations. First, a regularized learner estimates conditional outcome nuisance functions. Second, cross-fitting trains each nuisance function outside a respondent's held-out neighborhood fold. Third, an orthogonal score combines those out-of-fold predictions with randomized treatment variation. Orthogonality makes the estimating moment locally insensitive to small nuisance-estimation errors, while cross-fitting separates prediction from score evaluation and limits regularization and in-sample overfitting bias. These procedures improve covariate adjustment; they do not create identification or alter the experimental estimand.

4.2.1 Route A: Randomization-Aware AIPW/DML-IRM

Architecture and estimand. Route A is an augmented inverse-probability weighted estimator based on an interactive regression model (AIPW/DML-IRM). "Interactive" means that the outcome relationship may differ across assignment arms. For $d \in \{0, 1\}$, define the treatment-specific response function

$$\mu_d(x, s) = \mathbb{E}[Y_{ic} \mid D_c = d, X_{ic} = x, S_c = s]\,. \tag{6}$$

Rather than impose one additive campaign coefficient inside the nuisance regression, Route A estimates separate response surfaces for treated and control observations. If $k(i)$ denotes respondent i's held-out fold, the cross-fitted AIPW pseudo-outcome is

$$\begin{aligned}\widehat{\phi}_i^A &= \widehat{\mu}_{1,-k(i)}(X_{ic}, S_c) - \widehat{\mu}_{0,-k(i)}(X_{ic}, S_c)\\ &\quad + \frac{D_c\Big[Y_{ic} - \widehat{\mu}_{1,-k(i)}(X_{ic}, S_c)\Big]}{\widehat{p}_{S_c}} - \frac{(1 - D_c)\Big[Y_{ic} - \widehat{\mu}_{0,-k(i)}(X_{ic}, S_c)\Big]}{1 - \widehat{p}_{S_c}}.\end{aligned} \tag{7}$$

The first line is the model-based conditional treatment contrast. The second line corrects the treated and control response surfaces with inverse-design-probability-weighted outcome residuals.

The estimator averages the pseudo-outcomes,

$$\widehat{\theta}_A = \frac{1}{n}\sum_{i=1}^{n}\widehat{\phi}_i^A. \tag{8}$$

Thus Route A directly targets the respondent-weighted average ITT, $\theta_A = \mathbb{E}[Y(1) - Y(0)]$. It is a risk difference for binary endpoints and a mean difference for the participation index and cost outcomes.

Why this route is used. Route A is the preferred specification because its score matches the randomized design and the average ITT estimand. The assignment probability is supplied by the experiment, not selected by a treatment classifier, and separate μ_1 and μ_0 functions permit treatment-effect heterogeneity across observed covariates and strata. With a verified design propensity, the augmentation terms protect the estimating equation against first-order errors in the outcome fits. The learners therefore provide flexible regression adjustment and potential precision gains rather than a new source of causal identification. In this setting, it is more precise to emphasize orthogonal regression adjustment under a known propensity than to make a generic claim of double robustness.

DML implementation. For every outcome, the notebook first reduces the data to one record per neighborhood and constructs two folds. Entire neighborhoods are shuffled and allocated as evenly as support permits within each stratum-treatment-arm cell; no neighborhood is divided across folds. For each held-out fold and each arm d, the learner is trained only on training-fold neighborhoods with $D_c = d$. It then predicts both $\widehat{\mu}_0$ and $\widehat{\mu}_1$ for every respondent in the held-out neighborhoods. Equation (7) is evaluated only with these out-of-fold predictions. Sparse stratum-arm cells cannot always appear in both folds, so the notebook reports them and allows the nuisance predictions to pool partly through the other covariates; the known design probability continues to supply the treatment weights.

The prespecified final learner is elastic-net regression. Within each training fold, numeric features are median-imputed and standardized, while stratum is most-frequent-imputed and one-hot encoded. Elastic net solves a penalized least-squares problem of the form

$$\widehat{b} = \arg\min_b \left\{ \frac{1}{n_{\text{tr}}} \sum_{i\in\text{tr}} (Y_i - Z_i' b)^2 + \alpha \left[\lambda \|b\|_1 + \frac{1-\lambda}{2}\|b\|_2^2 \right] \right\}, \tag{9}$$

with $\alpha = 0.03$, $\lambda = 0.5$, and at most 20,000 iterations. Binary-outcome predictions are clipped to $[0, 1]$. The allowed nuisance features are baseline neighborhood measures of household size, expenditure, prior collector visits, prior property-tax payment, tax-ministry knowledge, political participation, trust in the tax ministry, and trust in government. Endline wealth and business ownership, as well as registration, payment, visits, and other post-assignment variables, are excluded. Ridge, elastic net, and shallow histogram gradient boosting across ten split seeds are reported as stability diagnostics; formal Route A inference uses the prespecified elastic net and primary two-fold split.

Analytic uncertainty aggregates the centered pseudo-outcome at the assignment level. Let

$$\widehat{u}_i^A = \widehat{\phi}_i^A - \widehat{\theta}_A, \qquad \widehat{U}_c^A = \sum_{i:c(i)=c} \widehat{u}_i^A. \tag{10}$$

With G neighborhoods, the implemented finite-cluster-corrected variance is

$$\widehat{\mathrm{Var}}(\widehat{\theta}_A) = \frac{G}{G-1} \frac{\sum_{c=1}^{G} (\widehat{U}_c^A)^2}{n^2}. \tag{11}$$

Analytic confidence intervals use a t_{G-1} reference distribution, so neither cross-fitting nor covariance estimation treats respondents as independently assigned.

4.2.2 Route B: Design-Propensity PLR-DML

Architecture and estimand. Route B uses a partially linear residual-on-residual score. Define the pooled conditional outcome nuisance and treatment nuisance as

$$\ell_0(x, s) = \mathbb{E}[Y_{ic} \mid X_{ic} = x, S_c = s], \qquad m_0(s) = \mathbb{E}[D_c \mid S_c = s] = p_s. \tag{12}$$

For held-out observations, Route B constructs

$$\widetilde{Y}_i = Y_{ic} - \widehat{\ell}_{-k(i)}(X_{ic}, S_c), \qquad \widetilde{D}_i = D_c - \widehat{p}_{S_c}. \tag{13}$$

The orthogonal PLR score is

$$\psi_i^B(\theta, \eta) = \widetilde{D}_i \left(\widetilde{Y}_i - \theta \widetilde{D}_i \right), \tag{14}$$

and setting its sample mean to zero yields

$$\widehat{\theta}_B = \frac{\sum_{i=1}^{n} \widetilde{D}_i \widetilde{Y}_i}{\sum_{i=1}^{n} \widetilde{D}_i^2}. \tag{15}$$

This is a no-intercept regression of the cross-fitted outcome residual on design-residualized assignment.

Under a constant additive campaign effect, θ_B is that common ITT coefficient. Under heterogeneous effects $\tau(X, S)$ and stratum-specific assignment probabilities, however, PLR generally targets the treatment-variance-weighted projection

$$\theta_B = \frac{\mathbb{E}[p(S)\{1 - p(S)\}\tau(X, S)]}{\mathbb{E}[p(S)\{1 - p(S)\}]}. \tag{16}$$

It equals the unweighted average ITT under a constant treatment effect, a common assignment probability, or other conditions that make these weights irrelevant. This qualification distinguishes the PLR coefficient from Route A's direct heterogeneous-effect average.

Why this route is used. Route B is a complementary sensitivity analysis. It uses a different orthogonal score and only one pooled outcome model, asking whether the substantive findings survive a simpler partially linear residualization strategy. A pooled nuisance can be more stable than two arm-specific response surfaces, but the final coefficient compresses campaign effects into one linear projection and is less naturally suited to unrestricted heterogeneity. The estimates reported below are close across the two scores: the Route A-Route B difference is below 0.0015 for each binary outcome, below 0.0005 for both cost measures, and 0.009 for the participation index. This agreement indicates that the conclusions are not driven by one particular score, subject to the estimand distinction in equation (16).

DML implementation. Route B uses the same outcome-specific two-fold neighborhood partition, with whole neighborhoods allocated within stratum-treatment-arm cells. In each training fold, one elastic-net regression is fitted on all treated and control training observations to estimate $\ell_0(X, S)$. The model uses the same eight verified baseline features, fold-specific imputation and standardization, one-hot-encoded stratum, $\alpha = 0.03$, an ℓ_1 ratio of 0.5, and a 20,000-iteration limit; binary predictions are clipped to $[0, 1]$. Predictions for equation (13) are made only for held-out neighborhoods. Treatment is residualized by the audited design probability, not by a learned classifier. The final reported PLR coefficient again uses the prespecified elastic net and primary split, while the other learners and ten split seeds are stability checks.

The stratum indicators are mandatory design features in the pooled outcome nuisance, but the implementation passes their one-hot encodings through the same penalized elastic-net model as the other transformed regressors. Route B is therefore a stratum-aware pooled outcome nuisance with exact design-propensity treatment residualization; it is not literal full-sample within-stratum demeaning or an unpenalized fixed-effect regression. DML enters through the cross-fitted estimate of ℓ_0 and the orthogonal moment in equation (14), while the treatment nuisance remains fixed by the experiment.

For clustered analytic inference, define

$$\widehat{e}_i^B = \widetilde{Y}_i - \widehat{\theta}_B \widetilde{D}_i, \qquad \widehat{U}_c^B = \sum_{i:c(i)=c} \widetilde{D}_i \widehat{e}_i^B. \tag{17}$$

The implemented sandwich variance is

$$\widehat{\mathrm{Var}}(\widehat{\theta}_B) = \frac{G}{G-1} \frac{\sum_{c=1}^{G} (\widehat{U}_c^B)^2}{\left(\sum_{i=1}^{n} \widetilde{D}_i^2\right)^2}, \tag{18}$$

with analytic confidence intervals based on $G - 1$ degrees of freedom.

Design-consistent randomization inference. Formal inference for both routes fixes the pre-specified learner and primary split; estimates from ten split seeds remain diagnostics and are not pooled into a new standard error. For each of $B = 5{,}000$ draws, the notebook permutes campaign assignment across neighborhoods within each stratum while preserving the stratum-specific treated count and reruns the complete estimator. Route A refits both arm-specific outcome nuisances, and Route B refits its pooled outcome nuisance. The two-sided Monte Carlo randomization p-value is

$$\widehat{p}_{\mathrm{RI}} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left\{\left|\widehat{\theta}^{(b)}\right| \geq \left|\widehat{\theta}^{\mathrm{obs}}\right|\right\}}{B+1}, \qquad B = 5{,}000. \tag{19}$$

This design-consistent Monte Carlo randomization inference aligns the reference distribution with neighborhood assignment within strata and includes nuisance refitting. It approximates, rather than exhaustively enumerates, the full randomization distribution.

4.3 Replicated OLS and DML Results

Table 8 compares the published estimates, the replicated OLS estimates, and the two DML routes. All seven DML estimates are positive. Under both methods, six of the seven outcomes have randomization-inference p-values below 0.05; evaluation submission is the sole exception. Most DML estimates are moderately smaller and less precise than the original OLS estimates, but the differences are not large enough to suggest a substantive contradiction.

Town hall attendance is essentially unchanged by DML, remaining particularly robust to both estimation strategies. Evaluation submission produces the weakest result. The effect falls about 0.4-0.6 points under DML with the confidence intervals including zero, and the RI p-values rising to .164 and .131. However, the estimates remain positive with their intervals including the OLS estimate. Similar rises in RI p-values are seen across outcomes, with all but Town hall attendance having higher standard errors. Despite this, signs, magnitudes, and the overall inferential pattern are highly consistent between OLS and DML. Is also important to note that the DML strategies use a more conservative adjustment set (removing select features from the original set) to protect against nuisance model estimation error. Naturally, the modest attenuation and larger DML standard erros reflect both the estimator and safer but less individually predictive feature set.

Table 8: Published OLS and DML re-estimates of Weigel Table IV

Outcome	N	Clusters	Control mean	(1) Table IV	(2) AIPW/IRM	(3) PLR
Panel A: Estimated treatment effects						
Town hall attendance	1,934	252	0.171	0.045	0.045	0.047
				[0.020]	[0.021]	[0.021]
Exact RI p-value				0.023	0.032	0.023
Evaluation submission	2,913	356	0.099	0.024	0.018	0.020
				[0.012]	[0.013]	[0.013]
Exact RI p-value				0.058	0.164	0.131
Town hall or evaluation	2,913	356	0.164	0.050	0.044	0.042
				[0.016]	[0.018]	[0.019]
Exact RI p-value				0.005	0.017	0.021
Town hall and evaluation	2,913	356	0.035	0.027	0.026	0.026
				[0.009]	[0.010]	[0.010]
Exact RI p-value				0.005	0.010	0.009
Participation index	2,913	356	-0.077	0.145	0.127	0.118
				[0.043]	[0.049]	[0.051]
Exact RI p-value				0.002	0.010	0.016
Cost participation, rel. w.	2,913	356	0.111	0.050	0.048	0.048
				[0.017]	[0.018]	[0.018]
Exact RI p-value				0.007	0.017	0.015
Cost participation 2, rel. w.	2,913	356	0.157	0.071	0.067	0.067
				[0.021]	[0.024]	[0.024]
Exact RI p-value				0.002	0.007	0.006
Learner	–	–	–	–	Elastic net	Elastic net
Cross-fitting folds	–	–	–	–	2	2
RI repetitions	–	–	–	–	5,000	5,000

Notes: The Published column reports the paper's Table IV estimates and randomization-inference p-values. Cluster-robust standard errors are in brackets. Rows labelled "Exact RI p-value" retain the original table terminology; the DML entries are Monte Carlo approximations based on 5,000 neighborhood-within-stratum permutations with nuisance refitting, not exhaustive enumeration. The DML columns use the prespecified elastic-net learner and the primary cluster-aware two-fold split. The known-propensity AIPW estimand remains identified, but nuisance predictions partly pool sparse stratum-arm cells.

4.4 Randomization Inference and Multiple Testing

Table 9 reports the multiple-testing adjustment for the two primary participation outcomes, town hall attendance and evaluation submission. This adjustment matters because the original paper treats these two outcomes as a primary family. The final DML p-values use design-consistent Monte Carlo randomization inference based on 5,000 neighborhood-within-stratum permutations with nuisance refitting, so the reference distribution reflects the experimental assignment mechanism rather than an individual-level iid approximation.

Table 9: Multiple-testing adjusted inference for the primary Table IV outcomes

Outcome	OLS RI p	OLS Sankoh p	AIPW RI p	AIPW Sankoh p	PLR RI p	PLR Sankoh p
Town hall attendance	0.029	0.043	0.032	0.048	0.023	0.035
Evaluation submission	0.056	0.084	0.164	0.239	0.131	0.192

Notes: Sankoh-adjusted p-values follow the notebook's correction for the two primary participation outcomes, town hall attendance and evaluation submission, using the estimated correlation $\rho = 0.398$ across the two outcomes. AIPW and PLR p-values are Monte Carlo randomization p-values based on 5,000 neighborhood-within-stratum permutations with nuisance refitting.

Town hall attendance remains statistically meaningful after the Sankoh adjustment. The adjusted p-value is 0.043 for replicated OLS, 0.048 for AIPW, and 0.035 for PLR. This reinforces the central participation finding because the outcome is both substantively important and robust to design-consistent adjusted inference.

Evaluation submission does not survive the same adjustment. The adjusted p-value is 0.084 for replicated OLS, 0.239 for AIPW, and 0.192 for PLR. The DML exercise therefore pushes the evaluation-submission result further into the suggestive category. The estimate remains positive, but the statistical evidence is not strong once flexible nuisance adjustment and adjusted inference are applied.

4.5 Conclusion

The DML re-estimation largely confirms Weigel's participation-dividend result. The strongest evidence is for town hall attendance, the composite participation outcomes, the participation index, and the cost participation measures. These outcomes remain positive under both DML routes, and their Monte Carlo randomization-inference p-values remain below conventional levels. The re-estimation therefore suggests that the original findings are not driven solely by conventional OLS adjustment.

At the same time, DML sharpens the outcome-specific interpretation. Evaluation submission alone becomes weaker. Its point estimate remains positive, which is consistent with the original direction, but the DML p-values are no longer close to conventional significance thresholds and the Sankoh-adjusted p-values are larger still. This means the participation dividend should not be summarized as equally strong across all individual components. The evidence is stronger for attending a town hall and for broader participation indices than for evaluation submission in isolation.

The design-aware DML exercise reinforces the core participation-dividend result while clarifying which components are robust. DML leaves the town hall effect essentially unchanged, modestly attenuates the composite and index effects, and weakens the stand-alone evaluation-submission result.

5 Conclusion

Design-aware DML provides a useful framework for robustness and diagnosis, but its value is conditional on the credibility of the original research design, the controls available, nuisance-model performance, overlap, and, in instrumental-variable settings, the stability of the residualized first stage. The evidence supports neither the claim that DML uniformly improves conventional estimates nor the claim that unchanged estimates imply that machine learning adds nothing. By combining orthogonal scores and cross-fitting (Chernozhukov et al., 2018), DML can reinforce conclusions through stability across functional forms, learners, scores, and splits, or qualify them by exposing learner-, fold-, first-stage-, denominator-, or outcome-specific sensitivity.

For Nunn and Wantchekon (2011), the partially linear re-estimates broadly preserve the negative relationship between ancestral slave-trade exposure and contemporary trust under stable regularized, ensemble, conservative, and cluster-aware specifications. The DML-IV evidence is more qualified: stable learners in the main run support the original direction, but weak residualized first stages and unstable score denominators under demanding cross-fitting and sensitivity checks make the IV re-estimation diagnostically fragile. For Depetris-Chauvin, Durante, and Campante (2020), the principal identity and intergroup-trust estimates remain close to the published fixed-effect estimates, suggesting that nonlinearities and interactions among observed respondent controls are unlikely to drive the findings once the event-timing fixed effects are preserved; this is functional-form robustness, not validation of the timing assumptions. For Weigel (2020), both design-aware DML routes largely reinforce the participation-dividend result for town-hall attendance, composite outcomes, the participation index, and cost measures, while the stand-alone evaluation-submission result weakens after flexible adjustment, randomization inference, and multiple-testing correction.

Three practical lessons follow. DML re-estimation should preserve the original source of identification, estimand, fixed effects or assignment mechanism, clustering level, and inferential design. Learners should be selected for nuisance performance and design-relevant diagnostics, never for a preferred treatment-effect sign, magnitude, or significance level. Repeated and cluster-aware cross-fitting, overlap and split-stability checks, and, where applicable, residualized first-stage and denominator diagnostics are therefore part of the empirical evidence rather than optional implementation details. The value added emphasized by Baiardi and Naghi (2024) can thus arise from either stability or an informative diagnosis of fragility.

The contribution is correspondingly limited: this paper evaluates robustness to flexible adjustment for observed covariates; it does not create a new identification strategy or establish that a DML estimate is closer to an unknowable causal truth. DML cannot resolve unobserved confounding, validate an exclusion restriction or instrument, prove that event timing is exogenous, replace random assignment or design-consistent randomization inference, or automatically compensate for weak overlap, few clusters, sparse treatment, or limited covariate information. Future re-estimation work should combine orthogonal machine-learning methods with research-design-specific cross-fitting and inference rather than treat DML as a generic plug-in estimator.

6 References

- Athey, Susan, and Guido W. Imbens. 2019. "Machine Learning Methods That Economists Should Know About." *Annual Review of Economics* 11: 685–725. `https://doi.org/10.1146/annurev-economics-080217-053433`.

- Baiardi, Anna, and Andrea A. Naghi. 2024. "The Value Added of Machine Learning to Causal Inference: Evidence from Revisited Studies." *The Econometrics Journal* 27(2): 213–234. `https://doi.org/10.1093/ectj/utae004`.

- Breiman, Leo. 2001. "Random Forests." *Machine Learning* 45(1): 5–32. `https://doi.org/10.1023/A:1010933404324`.

- Breiman, Leo, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. 1984. *Classification and Regression Trees*. Belmont, CA: Wadsworth International Group.

- Chen, Tianqi, and Carlos Guestrin. 2016. "XGBoost: A Scalable Tree Boosting System." In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. `https://doi.org/10.1145/2939672.2939785`.

- Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. "Double/Debiased Machine Learning for Treatment and Structural Parameters." *The Econometrics Journal* 21(1): C1–C68. `https://doi.org/10.1111/ectj.12097`.

- Depetris-Chauvin, Emilio, Ruben Durante, and Filipe Campante. 2020. "Building Nations through Shared Experiences: Evidence from African Football." *American Economic Review* 110(5): 1572–1602. `https://doi.org/10.1257/aer.20180805`.

- Dietterich, Thomas G. 2000. "Ensemble Methods in Machine Learning." In *Multiple Classifier Systems*, edited by Josef Kittler and Fabio Roli, 1–15. Lecture Notes in Computer Science 1857. Berlin: Springer. `https://doi.org/10.1007/3-540-45014-9_1`.

- Friedman, Jerome H. 2001. "Greedy Function Approximation: A Gradient Boosting Machine." *The Annals of Statistics* 29(5): 1189–1232. `https://doi.org/10.1214/aos/1013203451`.

- Nunn, Nathan, and Leonard Wantchekon. 2011. "The Slave Trade and the Origins of Mistrust in Africa." *American Economic Review* 101(7): 3221–3252. `https://doi.org/10.1257/aer.101.7.3221`.

- Rumelhart, David E., Geoffrey E. Hinton, and Ronald J. Williams. 1986. "Learning Representations by Back-Propagating Errors." *Nature* 323(6088): 533–536. `https://doi.org/10.1038/323533a0`.

- Tibshirani, Robert. 1996. "Regression Shrinkage and Selection via the Lasso." *Journal of the Royal Statistical Society: Series B (Methodological)* 58(1): 267–288. `https://doi.org/10.1111/j.2517-6161.1996.tb02080.x`.

- Wager, Stefan, and Susan Athey. 2018. "Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests." *Journal of the American Statistical Association* 113(523): 1228–1242. `https://doi.org/10.1080/01621459.2017.1319839`.

- Weigel, Jonathan L. 2020. "The Participation Dividend of Taxation: How Citizens in Congo Engage More with the State When It Tries to Tax Them." *The Quarterly Journal of Economics* 135(4): 1849–1903. `https://doi.org/10.1093/qje/qjaa019`.

- Zou, Hui, and Trevor Hastie. 2005. "Regularization and Variable Selection via the Elastic Net." *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 67(2): 301–320. `https://doi.org/10.1111/j.1467-9868.2005.00503.x`.